

# Ritisha Singh

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## SUMMARY

Backend engineer building distributed platform services in Java and Spring Boot at Goldman Sachs, in a regulated financial domain. Applied NLP depth from a year of transformer research: author of a fine-grained NER corpus and a ten-model benchmark reaching 97.8% macro F1. Seeking backend or applied-ML engineering roles.

## EXPERIENCE

### Software Development Engineer, Analyst – Goldman Sachs

Jul 2025 – Present

Asset Management Technology | Bengaluru, India

- Build **Java/Spring Boot** platform services that orchestrate reads and writes across a large number of internal systems with differing availability guarantees and refresh cadences.
- Keep distributed state consistent using **idempotent write paths, retry and compensation logic**, and scheduled reconciliation jobs over a **PostgreSQL** view; cache hot reference data in **Redis** to cut repeated cross-system lookups.
- Consolidated operational data from several disconnected systems into a single view and layered a **rules-driven exception engine** on top, surfacing data breaks and missing setup before they reach downstream teams.
- Built the operator-facing front end in **React**, owning the feature end to end from service layer to screen.

### Research Intern – Complex Systems Laboratory, IIIT-Delhi

Aug 2024 – May 2025

B.Tech thesis: deep learning for named entity recognition in recipes

- Built **RecipeNER-FG: 10,785 manually annotated ingredient phrases (85,920 tokens)** from RecipeDB, sampled via **spherical k-means** over TF-IDF for lexical diversity, under a fine-grained **8-tag schema**.
- Validated annotation quality on a 250-phrase doubly annotated subset at **Cohen's  $\kappa = 0.90$**  and **Krippendorff's  $\alpha = 0.90$** , with formal tag definitions and adjudicated boundary cases.
- Benchmarked **ten transformer models** (BERT, RoBERTa, DistilBERT, DeBERTa-v3, FoodBaseBERT, BERT+CRF); **DeBERTa-v3-large reached 97.82% macro F1** on a 3,235-phrase held-out set.
- Documented a reproducible **negative result**: WordNet synonym augmentation cost 0.18–0.89 macro F1 across all ten architectures, traced to disrupted distributional cues at entity boundaries.

### Teaching Assistant, Machine Learning (CSE543) – IIIT-Delhi

Aug 2024 – Dec 2024

Under Prof. Jainendra Shukla

- Supported **450+ students**: ran tutorials, resolved technical queries, and coordinated assignment design and grading.

### Research Intern – 80L Robotics (ARRT Lab)

Aug 2023 – May 2024

Unmanned aerial vehicle software | Delhi, India

- Established a **software-in-the-loop (SITL)** environment with **ArduPilot** and **MAVProxy** so UAV control software could be tested without flight hardware; shipped a stable **Android build of QGroundControl 5.15.2** and simplified the ground-control interface.

## PROJECTS

### Raft Consensus with Leader Lease – Python, gRPC, Protocol Buffers, GCP

[github.com/Ritisha-21089](https://github.com/Ritisha-21089)

- Implemented **Raft** with a **leader-lease modification** serving reads without a consensus round, on a multi-node cluster across GCP VMs; handled persistent logs, metadata, client GET/SET over **gRPC**, and **leader failover**.

### Decipher & Decide – T&C simplifier; Python, NLP, IR, RAG, Tesseract

[github.com/waningcrescent](https://github.com/waningcrescent)

- Evaluated **five language models** for privacy-policy summarisation and selected **Legal-Pegasus**; added multilingual translation, **OCR** for scanned policies, and a retrieval-backed chatbot over an indexed summary database.

## EDUCATION

### B.Tech, Computer Science and Engineering – IIIT-Delhi

2021 – 2025

CGPA 8.60/10 | final two semesters: SGPA 9.60 and 9.56

- **Coursework**: Distributed Systems, Computer Networks, Operating Systems, Algorithms, Database Systems, Advanced Programming, Machine Learning, NLP, Information Retrieval, Artificial Intelligence, Data Science.

## TECHNICAL SKILLS

**Languages**: Java, Python, C/C++, SQL, JavaScript, HTML/CSS

**Backend & Distributed**: Spring Boot, REST APIs, gRPC, Protocol Buffers, RabbitMQ, ZeroMQ, Redis, PostgreSQL, MySQL

**ML & NLP**: PyTorch, Hugging Face Transformers (BERT, RoBERTa, DeBERTa-v3), scikit-learn, NumPy, Pandas, TensorFlow

**Frontend, Cloud & Tools**: React, Amazon Web Services, Google Cloud Platform, Microsoft Azure, Git, GitLab, IntelliJ IDEA, Linux,  $\LaTeX$

**Certifications**: Microsoft Azure Fundamentals (AZ-900); Microsoft Azure AI Fundamentals (AI-900)